

Charotar University of Science and Technology
M.Sc. (Nano Science & Technology) Semester II Examination April. 2010
NT710 Nanocharacterization Techniques

Date: 30th April 2010

Time: 10.00 am to 10.30 am

Max. Marks 20

Part-A

1. State the following wavelength range of an electromagnetic spectrum: (3)
Ultraviolet long : _____
Ultraviolet short: _____
Visible : _____
Near Infrared : _____
Mid Infrared : _____
Far Infrared : _____
2. What kind of transition takes place in molecules when irradiated with infrared region of e. m. Spectrum (1)
 - a. Only rotational
 - b. Electronic accompanied by vibrational-rotational
 - c. Vibrational – rotational
 - d. All of above
3. Arrange the functional groups in decreasing order of force constant/absorption frequency (stretching mode in wave number) of double bond in groups given below (In general): (1)

>C=O
I

>C=C<
II

>C=N<
III

 - (a) I > III > II
 - (b) III > I > II
 - (c) II > III > I
 - (d) II > I > III
4. In general, arrange mode of vibrations in increasing order of absorption frequency (wave number) (1)

Bending
I

Symmetric stretching
II

Asymmetric stretching
III

- (a) I < II < III
- (b) III < II < I
- (c) II < I < III
- (d) I < III < II

5. State the difference between contact and non-contact mode in AFM. (2)
6. Give the mathematical statement of Beer's law. Identify each term. (2)
7. Which of the following instruments are used to know the size of nanostructures > 20 nm? (1)
- (a) SEM and TEM
 - (b) SEM only
 - (c) TEM only
 - (d) AFM, SEM and TEM
8. Why X-rays are used for structural determination of a crystal? (1)
11. If the FWHM of a prominent peak in an X-ray diffraction pattern is 0.5° and is observed at a Bragg angle of 30° , estimate the particle size of the crystal. (2)

12. Compare the performance of EDAX and AES.

(3)

13. State the limitations of scanning tunneling microscope.

[2]

14. Draw the schematic diagram representing "electron lens" used in TEM.

[1]

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Part B

(Write each section on separate answer book)

SECTION I

- Que.1 (a) Can we use light scattering to find out the lattice parameter of the crystal? Give reason. (2)
- (b) Calculate the lattice parameter for an FCC structure if the Bragg angle (2θ) for (311) and (440) plane is 35° and 62.4° . (6)
- (c) Write Bragg's condition in vector notation, and draw the diagram for the same. (2)
- OR**
- Que.1 (a) Why XRD lines get broader when particle size decreases? (4)
- (b) Calculate particle size for plane (311) if the full width at half maximum is 1.2° . The wavelength of x-ray radiation used is 0.1974 nm. (3)
- (c) Can we use X-ray to determine magnetic moment of particle? Give reason. (3)
- Que. 2 (a) What is the difference between Static and dynamic light scattering? (3)
- (b) How DLS is used to detect particle size? (3)
- (c) What is the finger print region in FTIR spectrogram? Give the range of frequencies for each region of the spectrogram. (2)
- (d) What kind of transition takes place in molecules when irradiated with infrared region of electromagnetic spectrum? (2)

SECTION II

- Que.3 (a) List various structural characterization techniques used for nano materials. (2)
- (b) With diagrams, explain the scanning electron microscopy (SEM) in detail. (6)
- (c) State its merits and demerits of SEM. (2)
- Que.4 (a) Explain the working principle of an atomic force microscope. State its advantages compared to STM. (5)
- (b) What are the three different modes of under which AFM works? Explain Tapping mode. (5)
- OR**
- Que.4 (a) Briefly explain the concept of Energy dispersive analysis of X-rays (EDAX). State its limitations. (5)
- (b) Name different methods used for chemical analysis of nano structured thin films. Describe the concept of XPS in detail. (5)
- Que.5 (a) Give schematic of TEM. Briefly explain the working of diffraction-mode TEM. (5)
- (b) Explain in brief basic principle of SPM. (3)
- (c) What precaution one has to take in order to achieve very high resolution in SPM. (2)
